

Diagram illustrating matrix multiplication $A \times b = c$.

Matrix A (3x3):

	0	1	2
0	0	0	3
1	1	0	0
2	0	2	4

Vector b (3x1):

0

Vector c (3x1):

0
1

The diagram shows the calculation of the first element of c (row 1, column 0 of A multiplied by the first element of b), which is 1. The result is highlighted in red.